

CHANDLER PHUNG

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SUMMARY

Graduate student pursuing MS in Computer Science (December 2026) with hands-on experience building and training deep learning models and deploying full-stack data-driven applications. Proficient in Python, PyTorch, and scikit-learn with project experience spanning NLP, computer vision, and time-series prediction across end-to-end ML pipelines.

EDUCATION

University of Alabama at Birmingham

Birmingham, AL

MS, Computer Science

Expected, December 2026

Relevant Coursework: Advanced Data Structures & Applications, Machine Learning, NLP (Natural Language Processing), GPU Programming, Deep Learning, Foundations of Data Science

BA, Computer Science

December, 2024

BS, Kinesiology with a concentration in exercise science

May, 2023

WORK EXPERIENCE

Operations Manager & Software Developer | Classic Nail Spa

January 2019 - Present

- Architected a full-stack internal business dashboard (Flask REST API + React + Recharts) visualizing live POS sales, employee revenue contribution, and shop KPIs across 6 dashboard views — reverse-engineered vendor SQLite schema without documentation to expose endpoints aggregating nested JSON transaction data across multiple tables
- Containerized and deployed full-stack application on Linux Linode server using Docker and Docker Compose; configured UFW firewall, CORS policies, and Flask-Login session authentication to secure access to live business data
- Built automated database sync pipeline using Python and Windows Task Scheduler that monitors POS SQLite file changes and transfers 40,000+ records to remote server via SCP — maintaining dashboard data within 10 seconds of live transactions
- Built automated SMS marketing pipeline in Python processing 1000+ customer records nightly via cron — queries POS SQLite DB for upcoming birthdays, dispatches personalized messages via Twilio API, and logs delivery status; deployed as persistent Windows service

TECHNICAL PROJECTS

Song2Color - Music Genre & Tag Color Visualization

March 2026 – April 2026

- Trained Word2Vec skip-gram models on Spotify co-occurrence data producing 32-dim genre and 64-dim tag embeddings; validated cluster quality with KNN coherence and UMAP visualization
- Designed VAE on 96-dim concatenated song vectors learning a structured 16-dim latent space enabling smooth interpolation between songs and color generation for unseen genre-tag combinations
- Mapped embeddings to HSV color space via PCA projection ensuring maximally contrasting colors for dissimilar genres; resolved posterior collapse via KL floor regularization and β -annealing

F1 Lap Time Prediction - LSTM Regression Model

March 2026 – April 2026

- Built end-to-end LSTM pipeline using FastF1 API on multi-season telemetry (speed, throttle, brake, RPM, gear, DRS) with distance-based resampling to produce fixed-length time-series inputs
- Integrated track embeddings and weather features for multi-modal prediction; implemented full ML pipeline including normalization, train/val/test splits, caching, and model checkpointing

TECHNICAL SKILLS

Programming Languages: Python, JavaScript/TypeScript, Java, C/C++, SQL, HTML/CSS

ML/AI: PyTorch, Scikit-learn, OpenCV, Word2Vec, VAE, LSTM, UMAP, PCA, Computer Vision

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Google Colab

Web & APIs: React, Node.js, Express.js, Flask, FastAPI, Angular, RESTful APIs

Cloud & DevOps: AWS (EC2, ALB, Route 53, ACM), Docker, Docker Compose, Linux, Git, OAuth 2.0

Databases: PostgreSQL, MongoDB, MySQL, SQLite, MongoDB Atlas